

Research Table - Name: Adam Attia - Field: Electrical engineering technology, automation and control

	Professional Standard <i>Code of conduct: Governmental Institution / Professional Organization</i>	Press Releases (two) <i>Case Report: Ethical Breach or Bill/Policy</i>	Unbowed <i>Memoir / Autobiography</i>
APA: Title, author(s)/organization & publication date	Ontario Association of Certified Engineering Technicians and Technologists. (2025, March). <i>Code of ethics and rules of professional conduct</i>. OACETT. https://www.oacett.org/getmedia/6186da57-e0a7-4e91-b3ca-1a6e3658b928/Code-of-Ethics-and-Rules-of-Professional-Conduct-update-March-2025.pdf Type: professional code of ethics (Ontario professional association created by the OACETT Act, 1998). Core anchors: public safety, integrity, competence.	#1 Canadian Centre for Cyber Security. (2025, October 29). <i>AL25-016: Internet-accessible industrial control systems (ICS) abused by hackers</i> [Alert]. Government of Canada. https://www.cyber.gc.ca/en/alerts-advisories/al25-016-internet-accessible-industrial-control-systems-ics-abused-hackers #2 Bronskill, J. (2025, October 30). Hacktivists targeting water, energy and food systems, Canadian security centre warns. <i>CBC News / The Canadian Press</i>. https://www.cbc.ca/news/politics/cyber-threat-water-energy-food-systems-9.6960580 Type: #1 = official government alert (primary source); #2 = news article reporting on #1 (secondary source).	Maathai, W. (2006). <i>Unbowed: A memoir</i>. Alfred A. Knopf. Chapters used: Chapter 6, "Foresters Without Diplomas" and Chapter 7, "Difficult Years". Author: Wangari Maathai (1940-2011), Kenyan biologist, founder of the Green Belt Movement (1977), Nobel Peace Prize 2004.
Summary: 1. <i>Outsourcing (articles): who's involved, what happened, when did it happen & how does it relate to your own duties</i> 2. <i>Unbowed: Choose an event that occurred in the chapters you have read that has impacted your perception of workplace values & responsibilities (Chapters 6-7 recommended)</i> • Describe it: Who, what, when & how	Who: OACETT, the association that certifies engineering technicians (C.Tech.) and technologists (C.E.T.) in Ontario. The code applies to every member. What: Two parts. (1) A Code of Ethics with 11 duties (a to k): hold paramount the safety, health and welfare of the public and the environment (a); accept assignments only when qualified by training or experience (b); give opinions only based on adequate knowledge and honest conviction (c); act with integrity, keep confidentiality, disclose conflicts of interest (d); keep informed to maintain competence (f); be fair and courteous, give credit where due (g); present clearly to the employer the consequences if a technical decision is overruled (h); report hazardous, illegal or unethical practices (i); act professionally on social media (k). (2) Rules of Professional Conduct that define misconduct: negligence (3.1), failing to safeguard life, health or property (3.2), failing to correct or report a	Who: The Canadian Centre for Cyber Security (Cyber Centre, a division of the Communications Security Establishment) and the RCMP, who published the alert; the victims: a water facility, a Canadian oil and gas company and a farm; the attackers: "hacktivists" (hackers acting for political or publicity motives). Article #2 was written by Jim Bronskill (The Canadian Press) and published by CBC. What: In "recent weeks" before October 29, 2025, hackers got into industrial control systems (ICS) that were connected directly to the internet. Three incidents: (1) water pressure valves were changed at a water facility, degrading service for the community; (2) an Automated Tank Gauge at an oil and gas company was manipulated, triggering false alarms; (3) temperature and humidity settings of a grain-drying silo were changed, creating potentially unsafe conditions. The exposed devices named are exactly my future tools: PLCs, RTUs, HMIs,	Event chosen: the "foresters without diplomas" (Chapter 6, 1977 onward). Who: Wangari Maathai, the National Council of Women of Kenya (NCWK), rural Kenyan women (mostly farmers with little or no formal schooling) and the professional foresters of the Kenyan Department of Forestry. What: After the first public tree planting on World Environment Day, June 5, 1977, in Kamukunji Park, Nairobi (the start of what became the Green Belt Movement), Maathai asked the government foresters to teach the women how to grow seedlings. The foresters explained that tree planting was a technical job that needed a diploma, with complicated methods the women could not follow. The women stopped listening to the experts, used their knowledge of the land, observed how seeds germinate naturally and learned by trial and error. Their nurseries worked, and Maathai started calling them "foresters without diplomas".

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	danger (3.3), not complying with codes and standards (3.4), signing a drawing you did not prepare or check (3.5), doing work you are not competent to do (3.8), hidden conflicts of interest (3.9), and the new rule on responsible, transparent use of AI tools (3.11). When: Latest update March 2025 (the social media and AI rules are the newest additions). Link to my duties: As an electrical / automation technician I will wire, program, test and maintain control panels, PLCs, HMIs and motor drives. The code tells me exactly what is expected: follow the Electrical Code and standards (3.4), never sign off a panel or drawing I did not actually check (3.5), refuse jobs I am not trained for, like high-voltage work (3.8), warn my supervisor in writing when a shortcut is dangerous (h, 3.6) and report unsafe installations even if it is uncomfortable (i, 3.3).	SCADA, Safety Instrumented Systems, Building Management Systems and IIoT devices. The attackers want media attention, to discredit organizations and to "undermine Canada's reputation"; organizations become "victims of opportunity". The alert recommends: inventory all internet-accessible ICS devices, remove direct exposure (VPN with two-factor authentication), add monitoring (intrusion prevention, penetration tests, vulnerability management), follow vendor hardening guidelines from installation to decommissioning, clarify roles and responsibilities, run tabletop exercises and report incidents to the Cyber Centre and the police. When: Alert published October 29, 2025; CBC article October 30, 2025. Link to my duties: The person who installs an HMI or a PLC and leaves it on a public IP address with a default password is often a technician, not a hacker. My duties will include configuring devices securely, applying vendor recommendations, documenting what is connected to the network and flagging exposed systems to my employer. The alert also says that unclear roles and responsibilities leave systems unprotected: knowing what I am responsible for is part of my job.	In the same chapter, the Department of Forestry first offered free seedlings and then withdrew the offer when the demand grew too fast, so the movement created its own community nurseries and paid the women a small amount for every seedling that was still alive after a few months (verified by Green Belt staff). Also (Chapter 7, 1979-1982): during her divorce, Maathai said publicly that the judge was either incompetent or corrupt; she was charged with contempt of court and jailed at Lang'ata Women's Prison for several days. In 1982 she resigned from the University of Nairobi to run for Parliament, was disqualified on a technicality and the university refused to take her back, so she lost her job and her housing but kept the Green Belt Movement alive. How it changed my perception: competence is proven by results and verification, not only by a title; experts can be wrong when they look down on people; and institutions (the Forestry Department, the court, the university) can fail you, so your personal values have to hold on their own.
Ethical, Legal and/or Social Implications • <i>Consequences (breaking professional ethical codes and/or personal code of conduct)</i>	Professional: breaking the code is "professional misconduct" (section 3). A complaint can lead to a discipline process, a reprimand, suspension or loss of the C.Tech. / C.E.T. certification, which means losing the right to use the title and, often, the job. Legal: negligence (3.1) or failing to safeguard life and property (3.2) can also mean civil liability (lawsuits for damages) and charges under safety laws (in Ontario, the Occupational Health and Safety Act and the Ontario Electrical Safety Code; in Quebec, the Loi sur la santé et la sécurité du travail and the Code de construction,	Ethical: the attackers put a community's water, a fuel site and a farm at risk to get attention: people are used as a means to send a message, which is exactly what deontological ethics forbids. On the other side, the owners and technicians who left devices exposed on the internet without protection failed in their duty of care (due diligence), even if nobody was hurt this time. Legal: for the attackers, the Criminal Code of Canada (unauthorized use of a computer, s. 342.1, and mischief in relation to computer data, s. 430(1.1)). For the	Ethical: the foresters' refusal to teach the women was a form of professional gatekeeping: using a diploma to exclude people instead of to protect the public. It is the opposite of the OACETT duty to "promote public knowledge" and to give credit where it is due. The Forestry Department withdrawing its promise of free seedlings shows how unreliable institutions push the cost onto the poorest. Legal: in Chapter 7, the divorce trial and the contempt-of-court charge show how the law can be used to silence a woman who criticizes authority; speaking the truth

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	chapitre Électricité). Signing a drawing you did not check (3.5) can make you personally responsible for a failure. Social: an electrical or control error can cause electrocution, fire, explosion or a machine that starts unexpectedly. Each accident reduces public trust in technicians and in the whole profession, which is why the code starts with the public, not with the client. Personal: the code also asks for honesty with yourself: admitting "I am not qualified for this" (b, 3.8) or "I did not check this" (3.5) is harder than staying quiet, but staying quiet is the real breach.	organizations, possible civil liability toward customers, and new obligations under Bill C-8 (2025), the proposed Critical Cyber Systems Protection Act, which would force federally regulated critical infrastructure operators to have cyber security programs and report incidents. The alert itself points out a legal gap: in sectors like water, food or manufacturing, "regulatory oversight does not cover cyber security". Social: loss of trust in municipal water and in the food supply; fear in small communities; small municipalities and family farms do not have cyber security budgets, so the burden falls on the technicians and service providers who install the systems. For my profession, the implication is that "it works" is no longer good enough: a system that works but is exposed is a hazard.	about a judge cost Maathai her freedom for a few days. Her disqualification from the 1982 election on a technicality and the university's refusal to reinstate her show the legal and economic consequences of challenging power in a one-party state. Social: rural women in 1970s Kenya had little access to land, education or money; deforestation meant walking farther for firewood and water and poorer nutrition for children. The Green Belt Movement turned an environmental problem into an issue of women's rights and democracy. For my own code of conduct: when a credential or a rule is used to look down on others rather than to protect them, it is being misused.
Evaluating sources: 1- Why is each source relevant? 2- What are your sources' strength and weaknesses? (e.g. credibility, intent & overall quality) 3- If you could create your own source, what would you do differently? What type of source would it be and what criteria would you abide by? (answer question 3 at the end, when and if you have time)	1- Relevance: it is the official duty-based (deontological) code for certified technicians and technologists, the exact profession I am studying for. Even in Quebec, where the equivalent is the Code de déontologie of the Ordre des technologues professionnels du Québec (OTPQ), the duties are almost identical, so it describes what will be expected of me. 2- Strengths: primary source, official, current (March 2025), written in clear rules ("shall") rather than vague guidelines; it even covers 2025 issues like social media and AI. Weaknesses: legalistic and dry, no examples or case studies, no explanation of how complaints are investigated or how often sanctions happen; it only applies to OACETT members in Ontario, so it has to be compared to the Quebec code. 3- My own source: see the answer in the Additional Notes row.	1- Relevance: both articles describe real 2025 incidents on the exact equipment of my program (PLCs, HMIs, SCADA, tank gauges, silo controllers) and show that a technical detail (a device left on the internet) becomes a public-safety and ethical problem. They also match the course: appeals (logos, ethos, kairos) and a possible slippery-slope fallacy. 2- Strengths of #1: primary source from a federal security agency (CSE) and the RCMP; concrete incidents and 13 precise mitigation steps; recent. Weaknesses of #1: written for CISOs and decision makers, not for the public; victims and attackers are anonymous, so the facts cannot be verified independently; the claim that incidents "undermine Canada's reputation" is not supported with evidence (slippery slope). Strengths of #2: credible public broadcaster, named journalist, Canadian Press standards, makes the alert accessible to everyone in one day. Weaknesses of #2: very short (about 1 minute to read); it mostly repeats the alert without	1- Relevance: a first-hand account of a professional woman who built an organization from nothing, dealt with experts, bureaucrats, courts and employers, and kept her values. It shows workplace values (competence, perseverance, accountability, respect for workers) in a context very different from mine, which makes them easier to see. 2- Strengths: credible author (first woman in East and Central Africa with a PhD, Nobel Peace Prize 2004), vivid detail, honest about her failures (Envirocare, the divorce, losing her job); strong ethos and pathos. Weaknesses: an autobiography gives one side only; it was written almost 30 years after the events, so memory and self-image shape the story; the foresters, the judge and the university never get to explain their point of view; the intent is partly to inspire, so it can idealize the movement. 3- My own source: see Additional Notes.

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		interviewing an expert, a victim or a municipality, so it adds no new information or verification. 3- My own source: see Additional Notes.	
Link to future workplace: <i>Identify values/duties/responsibilities involved</i> 1- <i>Explanation: use definitions by negation, by logical reasoning or textbook definitions</i> 2- <i>Visualisation: use real-world or hypothetical examples</i>	Values / duties: public safety (a), competence (b, f, 3.8), integrity and honesty (c, d), compliance with codes and standards (3.4), accountability for what I sign (3.5), duty to warn (h, 3.6) and duty to report (i, 3.3). 1- Explanation. <i>Competence</i> is not a diploma on the wall; by negation, it is the absence of work done beyond your training and experience. Logically, if public safety is paramount (a) and if I accept a job I cannot do safely (3.8), then I have already put the public second. <i>Accountability</i> is not blaming the drawing or the deadline; it is the fact that my signature means "I checked this" (3.5). <i>Duty to warn</i> means silence is not neutral: if my technical judgement is overruled and I say nothing, the code treats that as misconduct (3.6). 2- Visualisation. Real-world: an electrician is asked to add a circuit to a panel that is already over capacity; the client says "just make it fit". Under 3.4 and h, I must explain the consequences (overheating, fire), refuse the non-compliant solution and propose the right one. Hypothetical: my supervisor wants me to sign the commissioning report for a motor control center I did not test because the customer is waiting. Signing would be a breach of 3.5; the right action is to test it, even if it delays the project, or to let the person who tested it sign.	Values / duties: due diligence (vigilance), public safety, responsibility for essential services (water, energy, food), honesty about risks, confidentiality and clear roles. 1- Explanation. <i>Due diligence</i> is not "nothing bad has happened, so it is fine"; it is taking the precautions a reasonable and prudent practitioner would take before the incident (the exact definition of negligence in OACETT 3.1, reversed). Logically: an internet-accessible PLC with a default password can be modified by anyone; a water pressure setpoint controls public safety; therefore leaving that PLC exposed is a safety decision, not an IT detail. <i>Responsibility</i> for a control system does not end when the system works: the alert says devices must be secured "from deployment through decommissioning". 2- Visualisation. Real-world: a technician at a small municipal water plant discovers the HMI is reachable from the internet with the vendor's default login because the integrator set it up that way for remote support. The manager says "it has always worked". Following the alert and the code, I would document the exposure, propose a VPN with two-factor authentication, change the credentials and escalate in writing if nothing is done. Hypothetical: I install a grain-dryer controller on a farm and the owner wants to check it from his phone. Setting up a secure remote access and explaining why I refuse to open a port directly is part of my job, not an extra.	Values / duties: competence proven by results, respect for workers without credentials, verification and accountability, perseverance, integrity under pressure, honesty toward authority. 1- Explanation. <i>Competence</i> by negation: it is not the diploma; the diploma is proof that someone else trusts your competence, but the women proved competence by growing trees. A certification like C.Tech. or a Quebec electrician's licence matters because it protects the public, not because it makes me better than a worker without one. <i>Integrity</i> is not just not lying; by logical reasoning, if I only hold my values when they cost nothing, they are preferences, not values. Maathai held hers when they cost her a job, a house and a few days in prison. 2- Visualisation. Real-world: on a job site, an experienced electrician without a college diploma shows me a faster, safer way to pull cable; the OACETT duty to give credit where due (g) and Maathai's "foresters without diplomas" both say: listen, verify, and recognize him. Hypothetical: the Green Belt Movement paid women only for seedlings that were still alive after several months and checked by staff. That is exactly how commissioning works in automation: nobody gets paid for a system that "should" work; it is tested, verified and documented. If I ever manage a team, I would copy this: verify results, pay fairly, and never dismiss someone's knowledge because of their title.
Connection to broader contexts or other relevant issues <i>Connect each source to a broader context</i> • <i>socio-cultural, historical, political, ethical, etc.</i>	Historical / political: in Canada, professions are self-regulated under provincial laws (OACETT Act, 1998 in Ontario; the Code des professions and the OTPQ in Quebec), so the rules change from	International affairs: hacktivist attacks on water and energy systems are linked to the war in Ukraine and to tensions in the Middle East. In January 2024 a pro-Russia group made a water tower overflow in	Historical / political: post-independence Kenya (Jomo Kenyatta until 1978, then Daniel arap Moi), a one-party state where criticizing a court or running against the system had real consequences.

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• <i>current event(s), provincial elections, international affairs, etc.</i>	province to province even though technicians move across the country. In Ontario, the 2012 Algo Centre mall collapse in Elliot Lake, caused in part by a professional who signed off on an unsafe structure, pushed the engineering and technology professions toward stricter accountability, which is the spirit of rules 3.2 and 3.5. Current issues: the March 2025 update added rules on social media (k) and on AI (3.11): technicians now use AI tools to generate PLC code or diagnose faults, and the code says we remain accountable for the result. Ethical: this is a deontological standard in the Kantian sense: the duties hold regardless of the outcome; even if an uncertified installation never fails, doing it was still wrong.	Muleshoe, Texas; in November 2023 an Iran-linked group (CyberAv3ngers) took over Unitronics PLCs at a water authority in Aliquippa, Pennsylvania. Canada's National Cyber Threat Assessment 2025-2026 (cited in the alert) names critical infrastructure as a priority target. Political / legal: Bill C-8 (introduced June 2025) reintroduces the Critical Cyber Systems Protection Act after Bill C-26 died when Parliament was prorogued; the debate is about how far Ottawa should regulate private operators. Water, food and most manufacturing are provincial or municipal responsibilities, so the alert asks provinces to coordinate with municipalities: this becomes an issue for provincial elections (infrastructure budgets, Hydro-Québec's grid modernization, cyber security of municipal services). Socio-cultural / technological: IT and OT (operational technology) are converging; devices built 20 years ago to be isolated are now online; small municipalities and farms cannot afford security teams; AI lowers the skill needed to attack. Ethical: critical infrastructure is a public good: the Cyber Centre's "whole-of-society" approach is a call to shared responsibility, including the technician on the floor.	Deforestation came from colonial land policies, cash crops (tea, coffee) and population pressure; women, who fetched firewood and water, saw the effects first. Socio-cultural: the 1975 UN World Conference on Women in Mexico City and the UN Decade for Women (1976-1985) created the context in which the NCWK and the Green Belt Movement grew; Maathai's divorce trial ("too educated, too strong...") shows the patriarchal norms she was fighting at the same time. Ethical / international: the Nobel Peace Prize 2004 recognized the link between environment, democracy and peace. Today the same link appears in climate change (Canadian wildfires, COP negotiations) and in the debate about who pays for environmental protection. Link to my field: the energy transition (electrification, Hydro-Québec's 2035 plan, renewable grids) depends on technicians doing competent, honest work, the same values the Green Belt women showed with seedlings.
Additional Notes	Key quotations: "hold paramount the safety, health and welfare of the public" (2a); "undertake and accept responsibility of professional assignments only when qualified by training or experience" (2b); misconduct = "failure to act to correct or report a situation that the practitioner believes may endanger the safety or the welfare of the public" (3.3). Vocabulary: deontology = ethics of duty and rules (Kant), as opposed to consequentialism (judging by results). C.Tech. = Certified Technician; C.E.T. = Certified Engineering Technologist; OTPQ = Quebec equivalent (T.P. title).	Rhetorical appeals (from Research Task 2): logos (three concrete incidents, 13 mitigation steps), ethos (CSE, RCMP, Canadian Press), kairos ("in recent weeks", urgency), pathos ("potentially unsafe conditions"). Fallacy: possible slippery slope (small tampering "undermines Canada's reputation"; untargeted organizations become "victims of opportunity") without supporting evidence. Key quotations: "victims of opportunity"; "Unclear division of roles and responsibilities often creates gaps leaving critical systems unprotected"; "secure devices and services from deployment	Key quotations / ideas: "foresters without diplomas" (ch. 6); seedlings paid only when they survived (verification = accountability); the Forestry Department reversing its promise of free seedlings; the judge "incompetent or corrupt" and the contempt-of-court sentence (ch. 7); losing her university job and housing in 1982. Themes to use in the essay: competence vs. credentials; respect for workers; institutions that fail individuals; integrity that costs something; women's work as the base of environmental protection. Answer to evaluating question 3 (my own

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	Answer to evaluating question 3 (my own source): I would produce a case-study report based on interviews with Quebec technicians and OTPQ or OACETT discipline decisions: real anonymized cases, what rule was broken, what the sanction was and what the technician should have done. Criteria: primary interviews, cited rule numbers, publication dates, cross-checked with official decisions, reviewed by an instructor, and a clear statement of limits.	through decommissioning". Answer to evaluating question 3 (my own source): I would write an investigative article or technical report that follows one of the three incidents (for example the water facility) from installation to attack: interviews with the municipality, the integrator and the Cyber Centre, a timeline, the vendor's hardening checklist, the real cost of fixing it for a small town. Criteria: named sources where possible, verified facts, both sides heard (operator and security agency), accessible language for the public plus a technical appendix, and no fear-based claims without evidence.	source): I would create an oral-history collection (recorded interviews) with the Green Belt women themselves and with retired Kenyan foresters, to hear the people the memoir speaks about. Criteria: informed consent, several points of view (women, foresters, officials), dates and places checked against archives, transcripts published with the audio, and the author's own position stated openly instead of hidden.